ECON 470 Microeconomics of Artificial Intelligence

Prof. Eduardo Zambrano, Cal Poly

Course description. Microeconomic analysis of artificial intelligence. The economic value of AI prediction in online markets. AI's impact on consumer and firm decision-making. AI market automation including AI pricing. Policy issues including privacy, intellectual property, regulation, bias and fairness. Prerequisite: Consent of instructor. Recommended: ECON 3030 Intermediate Microeconomics.

Contact Information

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Learning Outcomes

What should students know or be able to do after taking this course?

1. Explain how AI prediction reduces uncertainty and creates economic value in decision-making contexts.
2. Analyze the demand for AI by identifying substitutes (rules, insurance, protection) and complements (judgment, data) to AI prediction.
3. Apply microeconomic models to evaluate when automation of decisions is optimal and how AI affects the allocation of decision authority.
4. Identify the economic incentives affecting AI data supply, including input data sharing and training data generation.
5. Compare AI pricing strategies and market structures, including versioning, bundling, and pricing to competitive versus monopolistic markets.
6. Analyze policy issues arising from AI, including market power, collusion, privacy, intellectual property, bias, and adoption regulation.
7. Synthesize course concepts by presenting and critiquing academic essays on the economics of transformative AI.

Course Logistics

Prerequisites

Consent of instructor. Recommended: ECON 3030 Intermediate Microeconomics (formerly ECON 311 Intermediate Microeconomics I).

Course Requirements

The course grade will be based on homework assignments [30%], class presentations [20%], a midterm exam [20%], and a final exam [30%].

- **Homework Assignments** (5 assignments, 30%): Problem sets applying microeconomic models to AI contexts. Homework assignments may be done in groups of up to two students but each student must write their answers in their own words and submit individually.
- **Class Presentations** (20%): Each student presents a Digitalist Papers essay and leads class discussion.
- **Midterm Exam** (20%): Covers Weeks 1–5 material.
- **Final Exam** (30%): Comprehensive, emphasizing Weeks 6–10 material.

Late work will be accepted at a discount. You lose 20% of the credit per calendar day late. The first 20% is lost immediately after an assignment is past its due date.

Time Management

This class will fully use all the time that you're theoretically expected to study for a 4-unit class, which is 8 to 12 hours per week, on average, across the quarter. I encourage you to keep track of the hours spent studying for this class. If you need help with time management and other study skills, take advantage of the resources offered by the Academic Skills Center: https://asc.calpoly.edu/ssl.

Academic Honesty

Learning cooperatively might seem to cloud the issue of academic honesty, but really it doesn't. The honesty issues relate to the group corporately. Directly copying homework answers is not cooperative learning. It is cheating.

Nobody begins the semester with the intention of cheating. Students who cheat do so because they fall behind gradually, and then panic at the last minute. Some students get into this situation because they are afraid of an unpleasant conversation with an instructor if they admit to not understanding something. I would much rather deal with your misunderstanding *early* than deal with its consequences later. **Please, do ask for help as soon as you feel you need it.**

- *Cheating* is defined as obtaining or attempting to obtain, or aiding another to obtain credit for work, or any improvement in evaluation of performance, by any dishonest or deceptive means. Cheating includes, but is not limited to: lying; copying from another's test or examination; discussion at any time of questions or answers on an examination or test, unless such discussion is specifically authorized by the instructor; taking or receiving copies of an exam without the permission of the instructor; using or displaying notes, "cheat sheets," or other information devices inappropriate to the prescribed test conditions; allowing someone other than the officially enrolled student to represent same.
- *Plagiarism* is defined as the act of using intentionally or unintentionally the ideas or work of another person or persons as if they were one's own without giving proper credit to the source. Such an act is not plagiarism if it is ascertained that the ideas were arrived through independent reasoning or logic, or where the thought or idea is common knowledge. Acknowledgement of an original author or source must be made through appropriate references, i.e., quotation marks, footnotes, or commentary.
- Do know that if I suspect cheating or plagiarism, I'm obligated to report the suspected incident to Cal Poly's *Office of Students Rights and Responsibilities*, and then they determine how to proceed, per their rules. See https://osrr.calpoly.edu/academic-integrity for details.
- On the other hand, cooperatively learning, and working collaboratively towards assignments is not only allowed but very much encouraged. All you have to do is to document the extent of the collaboration on your assignment "just as when citing any other source." If in doubt, talk to the instructor about this.

Disability Services

It is University policy to provide, on a flexible and individualized basis, reasonable accommodations to students who have disabilities that may affect their ability to participate in course activities or to meet course requirements. If you have a disability for which you are or may be requesting an accommodation, you are encouraged to contact both your instructor and the Disability Resource Center, https://drc.calpoly.edu/, Building 124, Room 119, (805) 756-1395, `drc@calpoly.edu` as early as possible in the term.

Diversity & Inclusion

Cal Poly considers the diversity of its students, faculty, and staff to be a strength and critical to its educational mission. Cal Poly expects every member of the university community to contribute to an inclusive and respectful culture for all in its classrooms, work environments, and at campus events. For more information on resources related to diversity and inclusion, please visit the Office of University Diversity & Inclusion website at http://diversity.calpoly.edu.

Student Privacy

If you have chosen to protect your Directory Information, it is important you communicate this to your instructor prior to or on the first day of class. This course uses course management tools that display students' full names and email addresses.

Technology Use

Electronic devices may only be used for specific learning activities. Use that inhibits the learning experience for you or others is prohibited. Please be considerate of those sitting around you and refrain from using the computer for non-class related matters. You can always get back to those later!

General Recommendations

1. Come to class (on time).
2. Complete the readings prior to lecture.
3. Complete and understand all homework assignments.
4. Discuss the course material with your classmates.
5. Do not wait until the night before homework is due to start your homework.
6. Do not be reluctant to ask for help (the professor and/or your classmates).
7. Do not wait to ask for help if you are feeling lost.

Readings

Required Text

Gans, Joshua (**G**) *The Microeconomics of Artificial Intelligence*, MIT Press, 2025. https://mitpress.mit.edu/9780262553544/the-microeconomics-of-artificial-intelligence/.

A PDF copy of the book will be freely available through MIT Press Direct Open Access.

Supplementary Readings

- Digitalist Papers (**DP**), Volume 2: "The Economics of Transformative AI." Stanford Digital Economy Lab, 2024. https://www.digitalistpapers.com/volume2.
- Agrawal, Ajay, Joshua Gans, and Avi Goldfarb. *Prediction Machines: The Simple Economics of Artificial Intelligence.* Harvard Business Review Press, 2018. https://www.predictionmachines.ai/.
- Karpathy, Andrej. `microgpt.py` — A complete GPT implementation in ~200 lines of pure Python with no dependencies. https://gist.github.com/karpathy/8627fe009c40f57531cb18360106ce95.

Expanded Course Content

Note: There will be homework assignments throughout the quarter. Student presentations on Digitalist Papers essays begin in Week 4.

Week 1: Introduction to AI and Economics

- The economic impact of AI: prediction, judgment, action, outcome
- Inside a GPT: guided walkthrough of Karpathy's `microgpt.py`
- How the model learns: backpropagation, the chain rule, and measuring uncertainty reduction
- What prediction can't do: breaking the model to reveal the limits of prediction and the role of judgment
- Readings: **G**, Chapters 1–2 (The Economic Impact of AI; Advances in Machine Learning). Karpathy, `microgpt.py`

Week 2: The Economic Value of Prediction

- The 2×2 model: value of perfect and imperfect information
- Biased signals and the ROC curve as a menu of error trade-offs
- The demand for prediction: heterogeneity in stakes and market demand
- Loss functions as acts of judgment, not prediction
- Readings: **G**, Chapter 3 (The Value of Prediction)

Week 3: Substitutes and Complements for Prediction

- Rules versus decisions
- Risk management activities: insurance and protection
- How prediction changes risk management: protection falls, insurance is ambiguous
- AI adoption decisions and the installed-base problem
- Asymmetric stakes: Amazon vs. Facebook
- Hidden uncertainty as a barrier to AI adoption
- Modeling judgment and its costs
- Complementarity between prediction and judgment
- Pre-thought judgment
- Readings: **G**, Chapter 4 (Substitutes for Prediction) and Chapter 5 (Complements to Prediction)
- **Homework 1 due**

Week 4: Automation and Decision Authority

- Tasks versus decisions
- Automation as delegation
- Machine substitution for skills
- Readings: **G**, Chapter 6 (Automation)
- Student presentations begin (Digitalist Papers essays)
- **Homework 2 due**

Week 5: System Effects and AI Supply

- AI adoption and organizational redesign
- The AI bullwhip effect
- Generation of input data and training data
- Readings: **G**, Chapters 7–9 (System Effects; Generation of Input Data; Generation of Training Data)

MIDTERM EXAM

Week 6: AI Pricing – Basic Models

- Market demand for prediction
- Monopolist supply of predictions
- Versioning and bundling

- Readings: **G**, Chapters 10–11 (Pricing with Exogenous Judgment; Pricing with Endogenous Judgment)
- **Homework 3 due**

Week 7: AI Pricing II – Market Applications

- Pricing to competitive markets
- Pricing to monopoly markets
- Prediction for negotiations
- Readings: **G**, Chapters 12–14 (Pricing to a Competitive Market; Pricing to a Monopoly Market; Prediction for Negotiations)

Week 8: AI Policy I – Competition and Collusion

- Market power in training data, input data, and predictions
- Data feedback loops
- AI and collusion: learning to collude
- Readings: **G**, Chapters 15–16 (Market Power; Collusion)
- **Homework 4 due**

Week 9: AI Policy II – Privacy, IP, and Misinformation

- The value of privacy and privacy regulation
- Intellectual property rights and AI-generated content
- Misinformation and content moderation
- Readings: **G**, Chapters 17–19 (Privacy Regulation; Intellectual Property Rights; Misinformation)

Week 10: AI Policy III – Bias, Regulation, and Social Impacts

- Bias and discrimination in AI
- Regulating AI prediction vs. regulating judgment
- Regulating adoption: reversible vs. irreversible
- Behavioral manipulation and changing social norms
- Readings: **G**, Chapters 20–22 (Bias and Discrimination; Regulating Adoption; Behavioral and Social Impacts)
- **Homework 5 due**

FINAL EXAM / FINAL PRESENTATIONS

Suggested Digitalist Papers Essays for Student Presentations

(from Volume 2: *The Economics of Transformative AI*, Stanford Digital Economy Lab)

1. Yoshua Bengio – Global catastrophic risks from AI (destructive chaos, concentration of power, loss of control)
2. Avital Balwit (Anthropic) – Preparing the next generation for transformed labor markets
3. David Autor & Neil Thompson – Labor scenarios from augmentation to full displacement
4. Ioana Marinescu – Adjustment insurance and digital dividends for workers
5. Anton Korinek & Lee Lockwood – Tax systems for the age of transformative AI
6. Ajay Agrawal & Joshua Gans – TAI as a "genius supply shock" and regulatory sandboxes
7. Joseph Stiglitz & Maxim Ventura-Bolet – AI, innovation, and misinformation
8. Betsey Stevenson – Meaning, purpose, and well-being when work is displaced
9. Anna Yelizarova – Global dividend mechanisms for sharing AI prosperity
10. Susan Athey & Fiona Scott Morton – Cheap Goods for Everyone? The Impact of Market Power in Artificial Intelligence on Welfare and Inequality